# Seeking dynamical models of cortical local field potentials via machine learning

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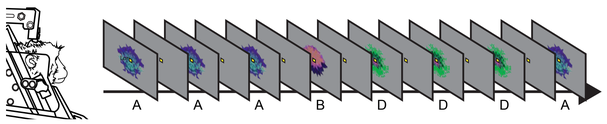
## Why model LFP dynamics?

- Local field potentials (LFPs) reflect synaptic and transmembrane currents from **local neuronal populaitions.**
- Clinical electrodes typically record population-level signals rather than single neurons; LFPs link animal and human recordings.
- Traditional analyses capture event-related responses and neural rhythms but **do not describe how LFP activity evolves over time**.
- A **governing equation** could enable simulation, prediction under untested conditions, and targeted experimental design.

Research Goal

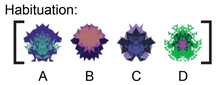
**Identify which aspects of LFP dynamics are robustly described by a governing equation across modeling choices.**

## Visual Behavioral Task




**Example Stimulus Pool**

Habituation:

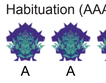



Deviants:

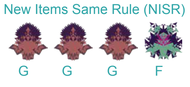



**Sequence Types**

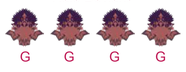
Habituation (AAAB or AAAA)




New Items Same Rule (NISR)




Rule Deviant




**Fixation Block**

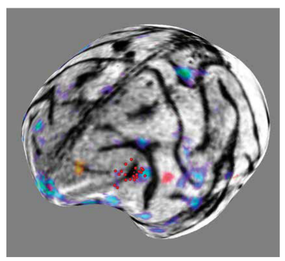
14.2 s

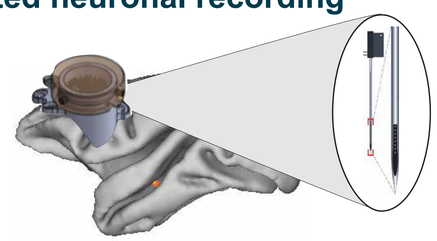
**Example Run**

= 1 four-item sequence




## Functionally-targeted neuronal recording

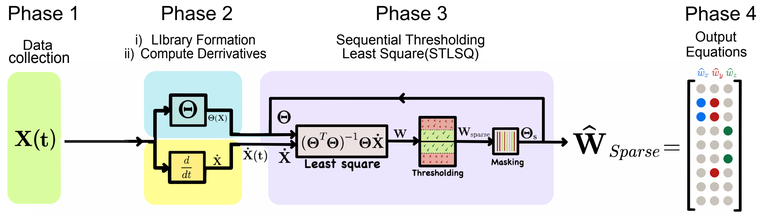




**Dataset**

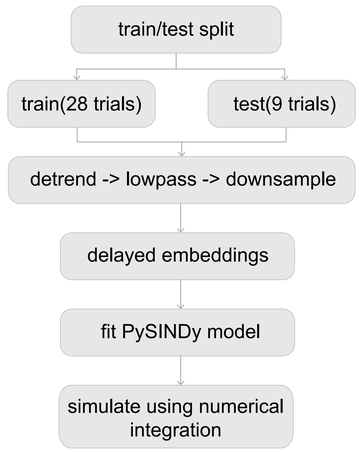
- 77 sessions = 38 Monkey W + 39 Monkey B
- 32 channel V-Probes, targeting dorsal sulcus.
- Single 16 or 32 channel V-Probe recordings
- Preliminary analysis: 1 session (Monkey B)
- Each sessions contains 1550 trial recordings

## SINDy algorithm




Sparse Identification of Nonlinear Dynamics(SINDy) was introduced in "Discovering governing equations from data by sparse identification of nonlinear dynamics systems"(Brunton et al., PNAS, 2016). Figure was adapted from the ngc-learn(https://ngc-learn.readthedocs.io/).

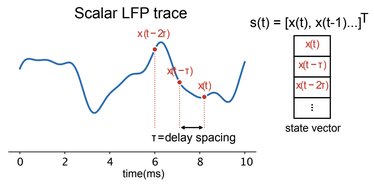
## Preprocessing Pipeline




**Hyperparameters**

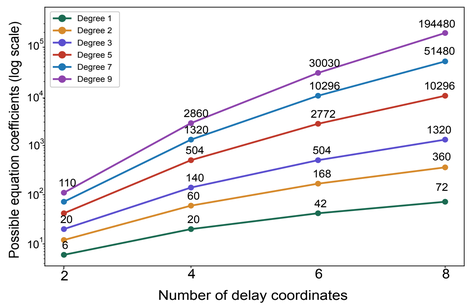
- **Polynomial degree** - library nonlinearity order
- **Low-pass cutoff** - retained frequency band
- **STLSQ threshold** - coefficient pruning cutoff
- **Number of delay coordinates** - embedded state size
- **Delay spacing** - steps between lags
- **Derivative smoothing window** - Savitzky-Golay span
- **Alpha** - ridge regularization strength

2075 configurations




Each scalar sample + its recent history → one state vector

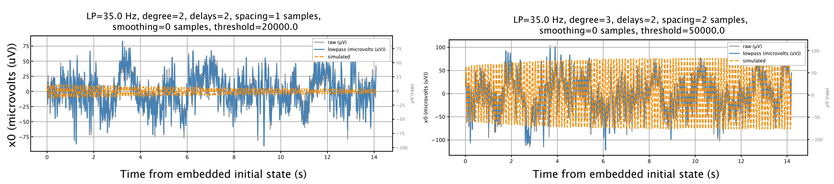
## Why Use Polynomial Library?



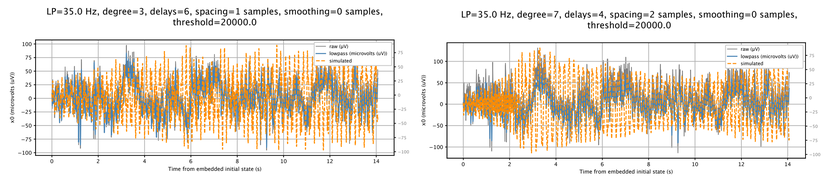

**Sustained oscillations require nonlinearity**, which low-order polynomial terms often capture. Many nonlinear systems have **purely polynomial dynamics**. With the governing dynamics unknown, polynomial libraries are SINDy's standard starting point, as they are **efficient and interpretable**.

## Three Regimes of Recovered Equations

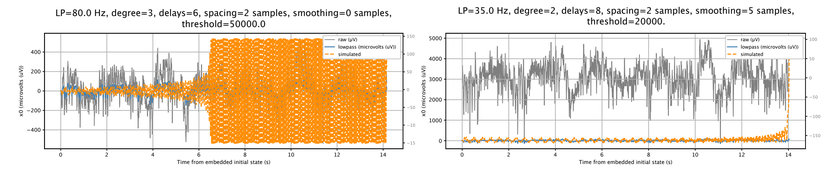
(i) fixed frequency, fixed amplitude oscillators




(ii) weakly nonlinear models with stable dominant frequency

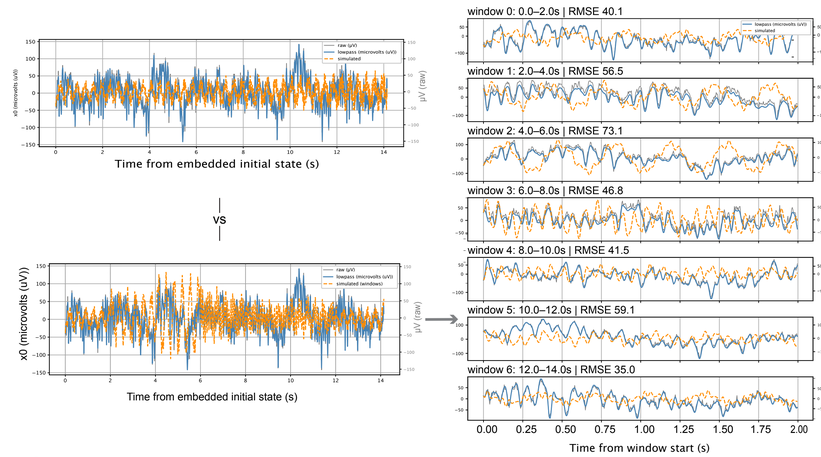



(iii) unstable models with significantly diverged trajectories




## Short vs. Long Prediction Horizons

LP=35.0 Hz, degree=2, delays=8, spacing=2 samples, smoothing=5 samples, threshold=20000.0




## Conclusions

- SINDy produced plausible dynamical models only under restricted conditions.
- Most hyperparameters produced models dominated by a fixed frequency oscillation, without capturing broader temporal structure of LFPs.
- Through hyperparameter sweeps, we distilled some interactions between hyperparameters, identifying configurations that tend to be more successful.

## Open Questions & Ongoing Work

- Could modified SINDy variants recover more realistic solutions?
- Does delay embedding representation itself drive the observed oscillatory dominance, rather than the signal?
- How might noise and latent inputs be incorporated?
- How would SINDy perform on multi-channel LFP recordings?
- Ongoing work is testing whether deep delay autoencoders (Bakarji et al., arXiv: 2201.05136) can improve on fixed delay embeddings.

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